

A HIERARCHICAL SALIENCY BASED TARGET DETECTION METHOD FOR HIGH-RESOLUTION SAR IMAGES

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ABSTRACT

The traditional target detection methods for high-resolution synthetic aperture radar (SAR) images rely on the high intensity contrast between the targets and clutter, which are usually effective on the homogeneous background; however, they may lose effectiveness in the heterogeneous background. Compared with the clutter, the targets of interest usually have specific size and shape characteristics in high-resolution SAR images. Based on what mentioned above, we propose a target detection method based on the hierarchical saliency (HS) for high-resolution SAR images. The proposed method constructs the Bayesian saliency map to obtain the complete structures of both the targets of interest and strong clutter with superpixels as units firstly. Then, the morphological saliency map is constructed to retain the targets of interest while suppressing the strong clutter via the introduction of prior size information. The experimental results on the MiniSAR images show the effectiveness of the proposed target detection method.

Index Terms— Synthetic aperture radar (SAR), target detection, saliency detection, hierarchical saliency (HS).

1. INTRODUCTION

Recently, the imaging technology of synthetic aperture radar (SAR) has been developed quickly. The amount of SAR images gradually increasing and the resolution becomes higher enough. However, the task of target detection still has some difficulties [1], [2] in complex high-resolution SAR images.

The two-parameter constant false alarm rate (CFAR) [3], a traditional target detection method for SAR images, assumes the background clutter obeying the Gaussian distribution for the pixel-level intensity differences between the targets of interest and clutter, of which the performance is good in the simple scenes. Considering that the pixel-level CFAR is easily effected by the speckle noise, Yu *et al.* [4] proposed a C¹FAR target detection method for SAR images using the modified simple linear iterative clustering (SLIC) to generate superpixels as units, and the superpixel-

based CFAR is robust to the speckle noise. However, the above mentioned CFAR have drawbacks. Firstly, choosing an appropriate clutter statistical model is difficult for real data, especially some complex scenes. Secondly, it is difficult to eliminate strong clutter false alarms with only considering the intensity information. The drawbacks mentioned above inspire us to develop an effective target detection method which does not need to model clutter statistical distribution and has less false alarms for SAR images.

Saliency detection has been widely developed in optical images recently. Numerous saliency detection methods for optical images have been proposed, e.g., Itti's method [5], spectral residual (SR) method [6]. Tong *et al.* [7] proposed the Bayesian saliency detection method. The method highlights the salient and suppress the non-salient regions via combining the prior map obtained from the superpixel-level local contrast method and the likelihood maps obtained from the convex hull within the Bayesian framework. However, due to the existence of speckle noise and the large SAR image scenes, it is difficult to obtain the convex hull for SAR target detection. Therefore, this method cannot directly be applied to the SAR images. Furthermore, the prior information of the targets is very useful for reducing false alarms, especially strong clutter.

Huang *et al.* [8], proposed a morphological saliency method based on differential morphological profiles (DMPs) for building extraction for optical remote sensing images. This method can highlight the targets while suppressing the backgrounds by make full use of prior information. However, due to the speckle noise, the intensities of the pixels in SAR images are fluctuant, making the morphological saliency map difficult to extract the targets via DMPs from the original SAR images.

According to the mentioned above, in this paper, we propose a hierarchical saliency (HS) based target detection method for high-resolution SAR images. The proposed method, firstly, constructs the Bayesian saliency map to obtain the complete structures of both the targets of interest and strong clutter with superpixels as units. Then, the morphological saliency map, is constructed by using DMPs on the Bayesian saliency map to retain the targets of interest while suppressing the strong clutter by the aid of the introduction of prior size information. The major contributions of this paper are summarized as follows.

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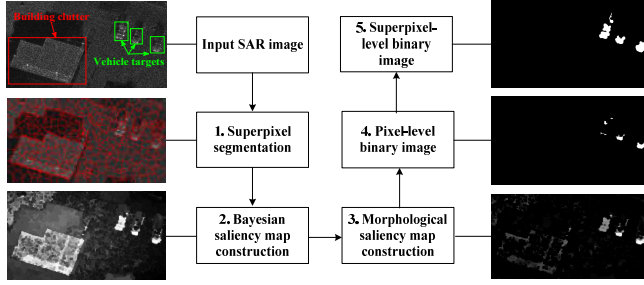


Fig. 1. Flowchart of the proposed target detection method.

- (1) We modified a Bayesian saliency detection method with the superpixels as basic units for SAR images, which can improve the completeness of both the targets and strong clutter and alleviate the effect of speckle noise.
- (2) We propose a hierarchical saliency method, in which the morphological saliency detection method is applied on the Bayesian saliency map to highlight the targets of interest while suppress the strong clutter for SAR images.

The remainder of this paper is organized as follows. Section 2 introduces the proposed HS method. Section 3 gives the experimental results on the MiniSAR images. Finally, Section 4 concludes this paper.

2. PROPOSED HS TARGET DETECTION METHOD

2.1. Framework of the proposed method

Fig. 1 shows the flowchart of the proposed HS target detection method. The brief introductions are given in the following.

Step 1: The input SAR image is segmented into the superpixels using the modified SLIC algorithm [4] which is more suitable for SAR images. In this paper, the superpixel is used as elementary unit to make the proposed HS method more robustness to speckle noise and more accuracy in preserving the shapes.

Step 2: The Bayesian saliency map is constructed to obtain the complete structures of both the targets and strong clutter via the Bayesian framework on the prior saliency map and the likelihood saliency maps.

Step 3: The morphological saliency map is constructed by using the morphological profiles on the Bayesian saliency map to highlight the targets while suppressing the strong clutter.

Step 4: From the morphological saliency map we can see that the intensities of the targets are much higher than those of the clutter. Hence a global threshold can be used on the morphological saliency map to generate the pixel-level binary image in which the targets be extracted.

Step 5: The superpixel-level binary image, i.e. the detection result, can generate from the pixel-level binary image through a threshold on the ratio of the number of

detected pixels in one superpixel to the amount of the pixels in it.

2.2. Bayesian saliency

2.2.1. Prior saliency map

After the segmentation step, an input SAR image is segmented into superpixels $\{s_i\}_{i=1}^N$, where N is the amount of superpixels. For each superpixel s_i , its prior saliency is defined as:

$$P(s_i) = \sum_{s_j \neq s_i} \omega_n(s_j) \cdot \omega_{sp}(s_j, s_i) \cdot d_l(s_j, s_i) \quad (1)$$

where $\omega_n(s_j) = \text{Num}(s_j)$ is the size weight which represents the amount of pixels in superpixel s_j ,

$\omega_{sp}(s_j, s_i) = \exp\left(-\frac{d_{sp}(s_j, s_i)}{\sigma^2}\right)$ represents the spatial

distance weight, $d_{sp}(\square\square)$ indicates the spatial Euclidean distance between two superpixels, σ controls its strength, $d_l(\square\square)$ indicates the spatial Euclidean distance between the intensity mean values of two superpixels.

Then, we normalize $P(s_i)$ as:

$$\bar{P}(s_i) = \frac{P(s_i)}{\sum_{i=1}^N P(s_i)} \quad (2)$$

Here the normalized prior saliency $\bar{P}(s_i)$ denotes the prior probability of the superpixel s_i belonging to the potential target region (foreground).

2.2.2. Likelihood saliency maps

Through the multi-scale target detection method in [9], we can obtain the preliminary foreground and background. And based on it, the intensity histograms of the foreground region and background region, H^1 and H^0 , can be obtained respectively. Then, the likelihood of one pixel z belonging to the foreground $L(z|S^1)$ and the background $L(z|S^0)$ can be defined as:

$$L(z|S^1) = \frac{N^1(z)}{N_{S^1}} \quad (3)$$

$$L(z|S^0) = \frac{N^0(z)}{N_{S^0}} \quad (4)$$

where $N^1(z)$ and $N^0(z)$ represent the amount of pixels with the same intensity value as z in H^1 and H^0 respectively, N_{S^1} and N_{S^0} denote the total pixel numbers of the foreground and background respectively.

2.2.3. Bayesian saliency map

The prior saliency map and the likelihood saliency maps are combined within the Bayesian framework to obtain the Bayesian saliency map. The Bayesian saliency of each pixel z is defined as:

$$B(z) = \frac{\bar{P}(z)L(z|S^1)}{\bar{P}(z)L(z|S^1) + (1 - \bar{P}(z))L(z|S^0)} \quad (5)$$

where $\bar{P}(z) = \bar{P}(s_i)$ with the pixel z belonging to the superpixel s_i .

2.3. Morphological saliency

2.3.1. DMP

The DMP along direction dir is calculated as:

$$MP(dir, L) = open^{re}(dir, L, B) \quad (6)$$

$$DMP(dir) = |MP(dir, L_{max}) - MP(dir, L_{min})| \quad (7)$$

where $open^{re}(\cdot, \cdot, \cdot)$ is the opening-by-reconstruction morphological filter, dir and L are the direction and the size of the linear structural elements (SEs) respectively, B is the Bayesian saliency map, $MP(dir, L)$ represents the morphological profiles (MP) of the opening-by-reconstruction with the linear SEs, L_{max} is the maximum size of the vehicle targets, and L_{min} is the minimum size of the vehicle targets.

The MP can preserve the objects with the sizes equal to or larger than L while remove the objects of which the sizes is smaller than L along the direction dir . Therefore, the difference between $MP(dir, L_{max})$ and $MP(dir, L_{min})$, i.e. $DMP(dir)$, can preserve the vehicle targets whose sizes are between L_{min} and L_{max} along the direction dir , while removing the others.

Here, the linear SEs consider both the size prior information and shape prior information. Therefore, it is appropriate for the target detection in SAR images in which the targets are vehicles with a limited size range and rectangular shape.

2.2.2. Morphological saliency map

Firstly, we combine the DMPs with different directions to make the saliency map more robust. Then, the morphological saliency map can be normalized as:

$$M_s = \frac{\sum_{dir} DMP(dir)}{N_{dir}} \quad (8)$$

where N_{dir} denotes the number of directions.

In this paper, we adopt a hierarchical framework, in which we firstly construct the Bayesian saliency map, and then based on the Bayesian saliency map we construct the morphological saliency map. Hence, we refer the final

saliency map as the HS saliency map.

3. EXPERIMENTAL RESULTS AND ANALYSIS

The miniSAR real images are acquired by Sandia National Laboratories of America. We select two typical complex SAR images with the resolution $0.1m \times 0.1m$ and the size 1638×2510 pixels from the miniSAR real dataset for our experiments, and refer to them as Image 1 and Image 2 respectively. Fig. 2 shows the two original SAR images and their superpixel-level ground truths, where the vehicles are the targets of interest and the others are regarded as the clutter. From Fig. 2 we can see that, besides vehicle targets, there are buildings strong clutter which are easily affect detection performance.

In the proposed HS method, L_{max} is set to 40 which considers the maximum size of the real target to be detected; L_{min} is set to 4 which should be smaller than L_{max} and larger than the possible small clutter. And four directions, i.e., $dir \in \{0^\circ, 45^\circ, 90^\circ, 145^\circ\}$, are considered.

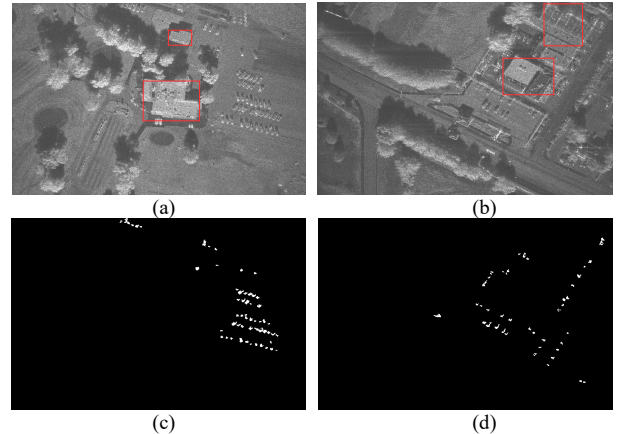


Fig. 2. Original SAR images and their ground truths. (a) and (c) are Image 1 and its superpixel-level ground truths of the vehicle targets, respectively. (b) and (d) are the Image 2 and its superpixel-level ground truths.

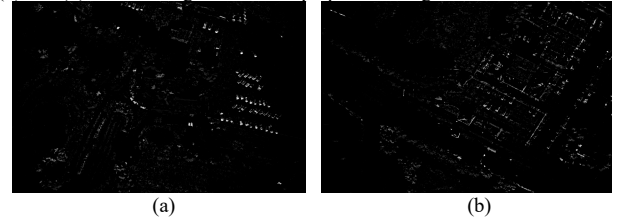


Fig. 3. (a) HS maps of Image 1, (b) HS maps of Image 2.

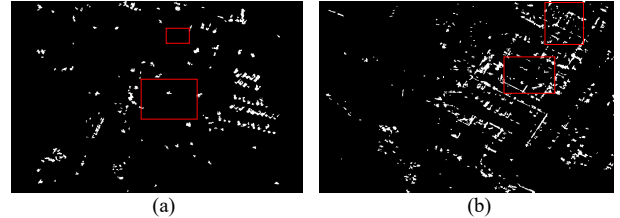


Fig. 4. (a) HS detection results of Image 1, (b) HS detection results of Image 2.

TABLE 1 AUCs OF DIFFERENT TARGET DETECTION METHODS AT THE SUPERPIXEL-LEVEL

Image	Superpixel-CFAR	Global-CFAR	Local-CFAR	PR	BS	MS	BMS
Image 1	0.9850	0.9744	0.9927	0.9823	0.9645	0.9855	0.9975
Image 2	0.9572	0.9504	0.9580	0.9684	0.9460	0.9610	0.9710

3.1. Saliency map results

Fig. 3 shows the HS map of Image 1 and the HS map of Image 2. From Fig. 3 we can see that the vehicle targets are much brighter than most the clutter. Therefore, we can detect the vehicle targets easily from the HS map with less false alarms.

3.2. Detection results

From Fig. 4 we can see the proposed method can remove much more building clutter. We compare the proposed HS method with two parameter CFAR (i.e., Local-CFAR [3], Global-CFAR [10], Superpixel-CFAR [4], PR [11], BS (Bayesian saliency map on original SAR images), and MS (morphological saliency map on original SAR images) in terms of the receiver operating characteristic (ROC) curve in Fig.5 and the area under the ROC curve (AUC) in Table 1 for quantitative evaluation.

As shown in Fig. 5, the ROC curves of the proposed HS method are better compared with those of the other target detection method. The higher the AUC value is, the better the performance is. From Table 1 we can clearly see that the proposed HS method has the highest AUC value than those of the other methods. Compared with BS method, the proposed method which adopts hierarchical structure can highlight targets while suppress clutter. And compared with MS method, the proposed method applies morphological saliency on the Bayesian saliency map is more suitable for SAR images.

4. CONCLUSION

To overcome the defects of the conventional SAR target detection methods, in this paper, we propose a new target detection method, called the HS method, for high-resolution SAR images. The proposed HS method, which hierarchically constructed Bayesian saliency map and morphological saliency map, make full use of the prior size information and prior shape information of the targets to highlight the targets and suppress the strong clutter. At the same time, by introducing the superpixel as unit, the proposed HS method alleviate the effect of the speckle noise, which is important for achieving better performance. The experimental results based on the MiniSAR real images have shown that the proposed HS method has the better performance for SAR target detection than some traditional target detection method and saliency detection methods.

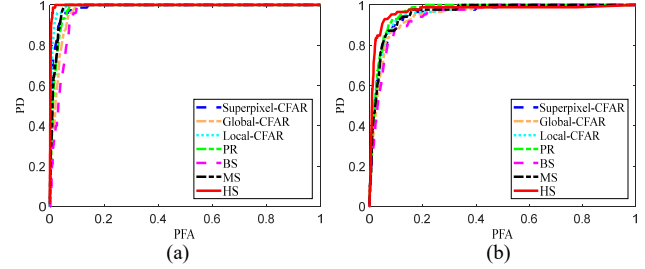


Fig. 5. (a) and (b) are the ROC curves of different target detection methods at the superpixel-level on Image 1 and Image 2 respectively. Here the red line represents the proposed method.

5. REFERENCES

- [1] D. E. Dudgeon and R. T. Lacoss, "An overview of automatic target recognition," *Linc. Lab. J.*, vol. 6, no. 1, pp. 3–10, 1993.
- [2] L. M. Novak, S. D. Halversen, G. J. Owirka, and M. Hiett, "Effects of polarization and resolution on the performance of a SAR automatic target recognition system," *Lincoln Lab. J.*, vol. 8, no. 1, pp. 49–68, 1995.
- [3] L. M. Novak, M. C. Burl, and W. W. Irving, "Optimal polarimetric processing for enhanced target detection," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 29, no. 1, pp. 234–244, Jan. 1993.
- [4] W. Y. Yu, Y. H. Wang, H. W. Liu, and J. L. He, "Superpixel-based CFAR target detection for high-resolution SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 5, pp. 730–734, May 2016.
- [5] L. Itti, C. Koch, and E. Niebur, "A model of saliency-based visual attention for rapid scene analysis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 20, no. 11, pp. 1254–1259, Nov. 1998.
- [6] X. D. Hou and L. Q. Zhang, "Saliency detection: a spectral residual approach," *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, pp. 1–8, 2007.
- [7] N. Tong, H. C. Lu, L. H. Zhang, and X. Ruan, "Saliency detection with multi-scale Superpixels," *IEEE Signal Processing Letters*, vol. 21, no. 9, pp. 1035–1039, Sep. 2014.
- [8] X. Huang and L. P. Zhang, "Morphological building/shadow index for building extraction from high-resolution imagery over urban areas," *IEEE J. Sel. Topics. Appl. Earth Observ. Remote Sens. (JSTARS)*, vol. 5, no. 1, pp. 161–172, Feb. 2012.
- [9] Z. C. Wang, L. Du, F. Wang, H. T. Su, and Y. Zhou, "Multi-scale target detection in SAR image based on visual attention model," *Proc. IEEE 5th Asia-Pacific Conference on Synthetic Aperture Radar (APSAR)*, pp. 704–709, Sep. 2015.
- [10] G. Gao, G. Kuang, Q. Zhang, and D. Li, "Fast detecting and locating groups of targets in high-resolution SAR images," *Pattern Recognit.*, vol. 40, no. 4, pp. 1378–1384, Apr. 2007.
- [11] H. P. Wang, F. Xu, and S. S. Chen, "Saliency detector for SAR images based on pattern recurrence," *IEEE J. Sel. Topics. Appl. Earth Observ. Remote Sens. (JSTARS)*, vol. 9, no. 7, pp. 2891–2900, Jul. 2016.